



Crossplatform architecture

Cross-platform
platform

www.libretro.com



RetroArch

● RetroArch

- A cross-platform architecture
- The reference frontend to an API
- An app library/ecosystem of its own
- A no-strings-attached enduser program
- A project with multiple stakeholders
 - Open source development community
 - Hardware vendors
 - Allied open source projects (XBMC, OpenEmu)



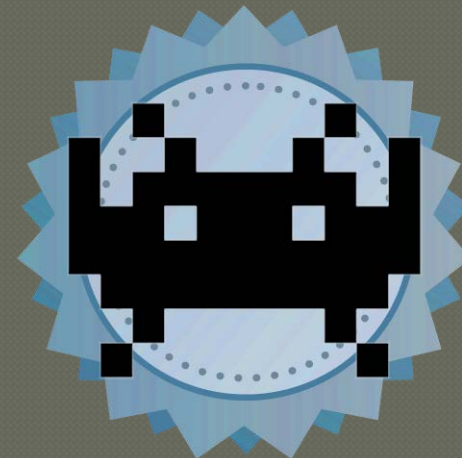
RetroArch

● RetroArch

- Goals/ambition
 - Create an all-encompassing platform on top of all existing platforms/operating systems/ecosystems
 - Any time, any place, any device
 - Backend and frontend compartmentalization
 - 1 Codebase per app

● Libretro

- A lightweight C/C++ API
- Works according to backend/frontend principle
 - Backend application talks with frontend by way of API
- Facilitates input/audio/video/camera/location streams
 - for use in games/media players/graphics applications
 - Augmented reality





Libretro

- ◉ Runs on nearly all available platforms/operating systems
 - Apps run on every platform
 - With added value (shaders, rewinding, input support, etc)
 - Are loaded in as module inside the frontend app
 - Dynamic library (Position Independent Code)
 - No recompilation of the frontend (RetroArch) necessary



Libretro

Dynamic or static library





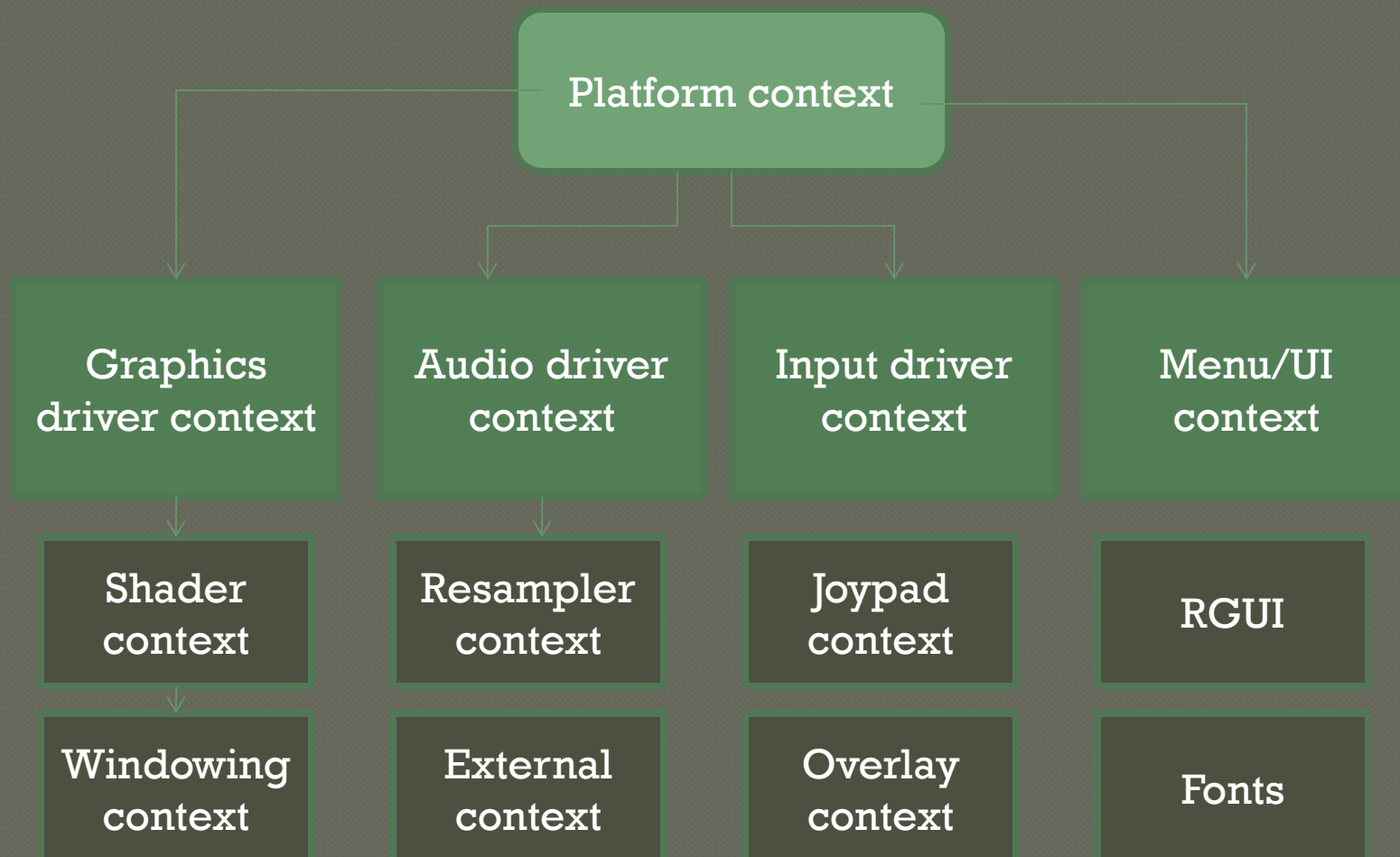
RetroArch

◉ Libretro frontend

- Reference implementation
 - RetroArch
 - Platform agnostic
 - Performance-focused
 - Advanced features (shaders, overlays, rollback, etc)
 - Implements new features of the API first
 - Puts a heavy emphasis on performance and code clarity
- Other implementations
 - XBMC (RetroPlayer)
 - Arcan
 - Minir (upcoming)

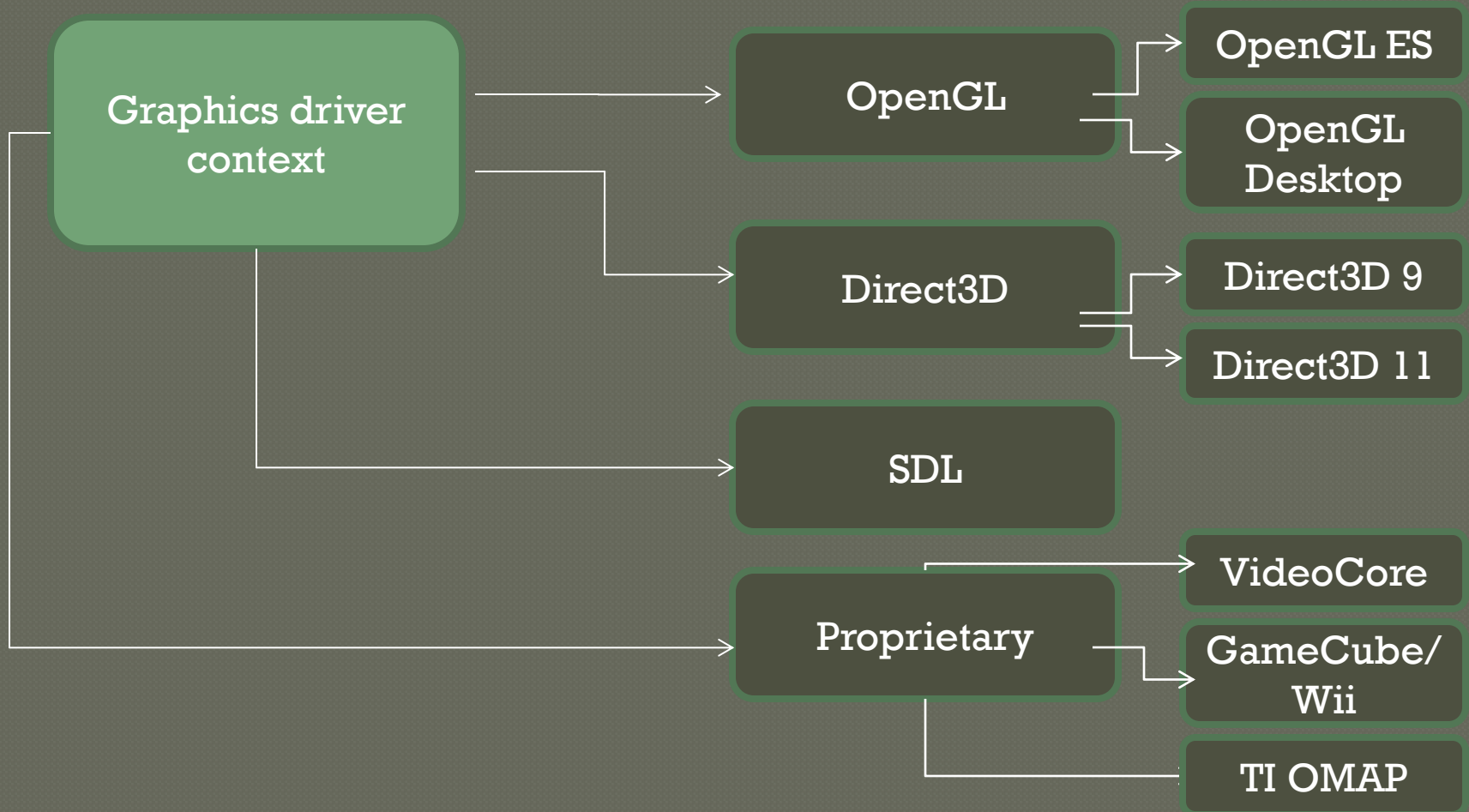


RetroArch



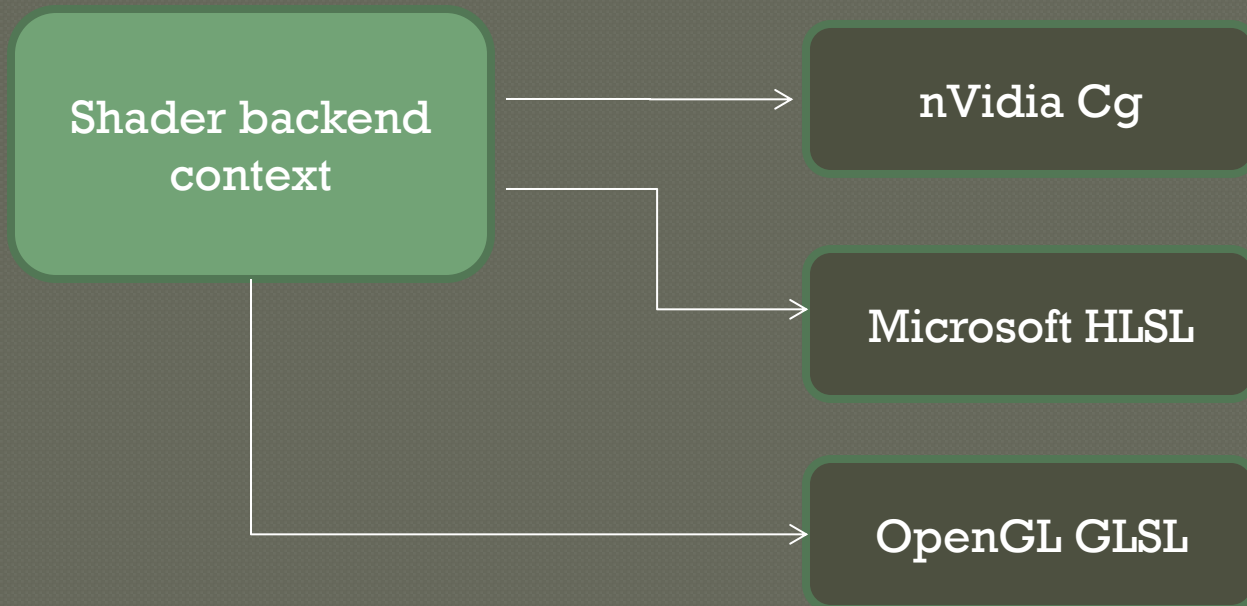


RetroArch





RetroArch

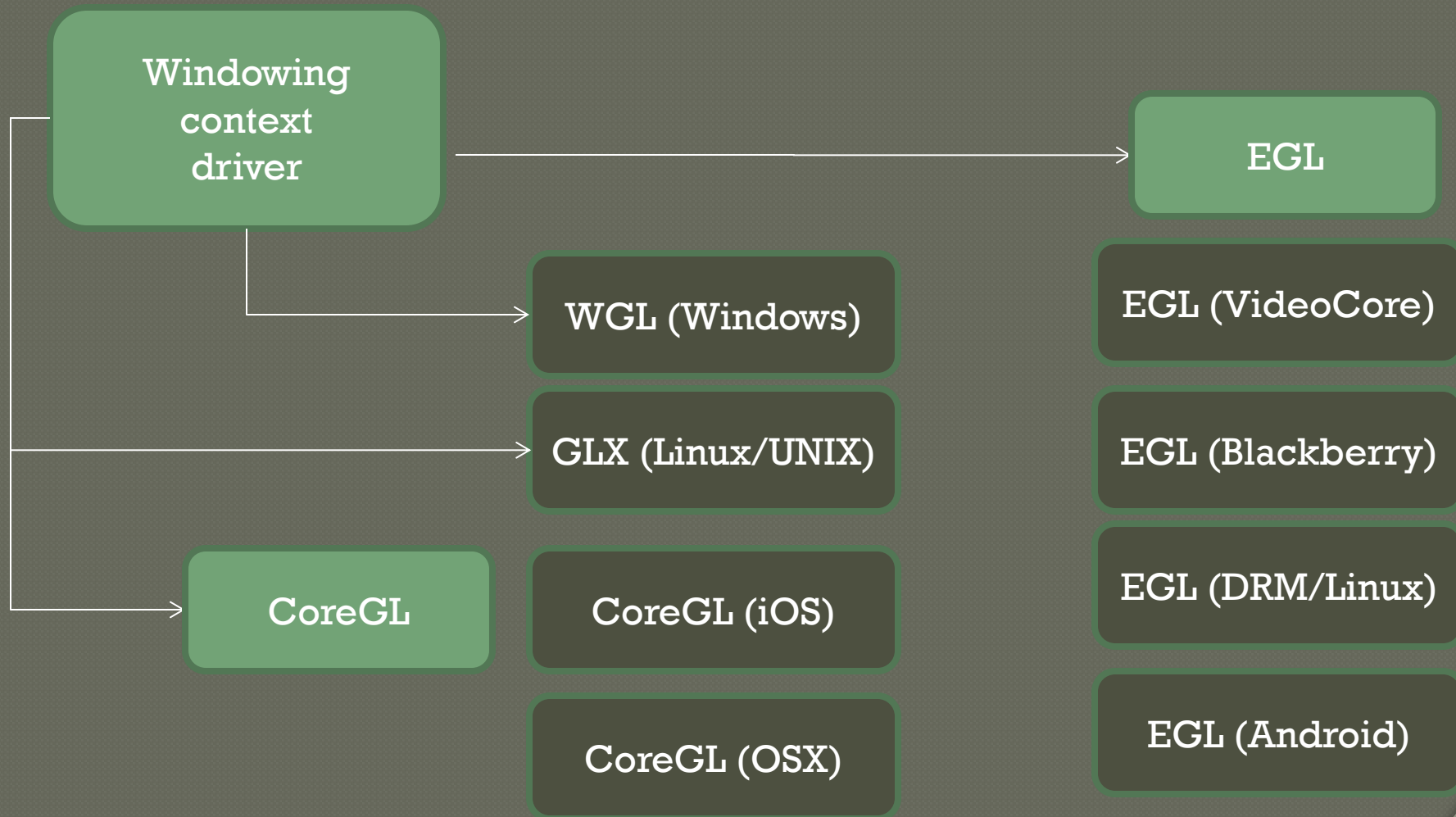


● Shaders

- Three totally different implementations
 - 1 Codebase
 - Can convert Cg to GLSL through off-line conversion script
 - Cg maps mostly to raw HLSL

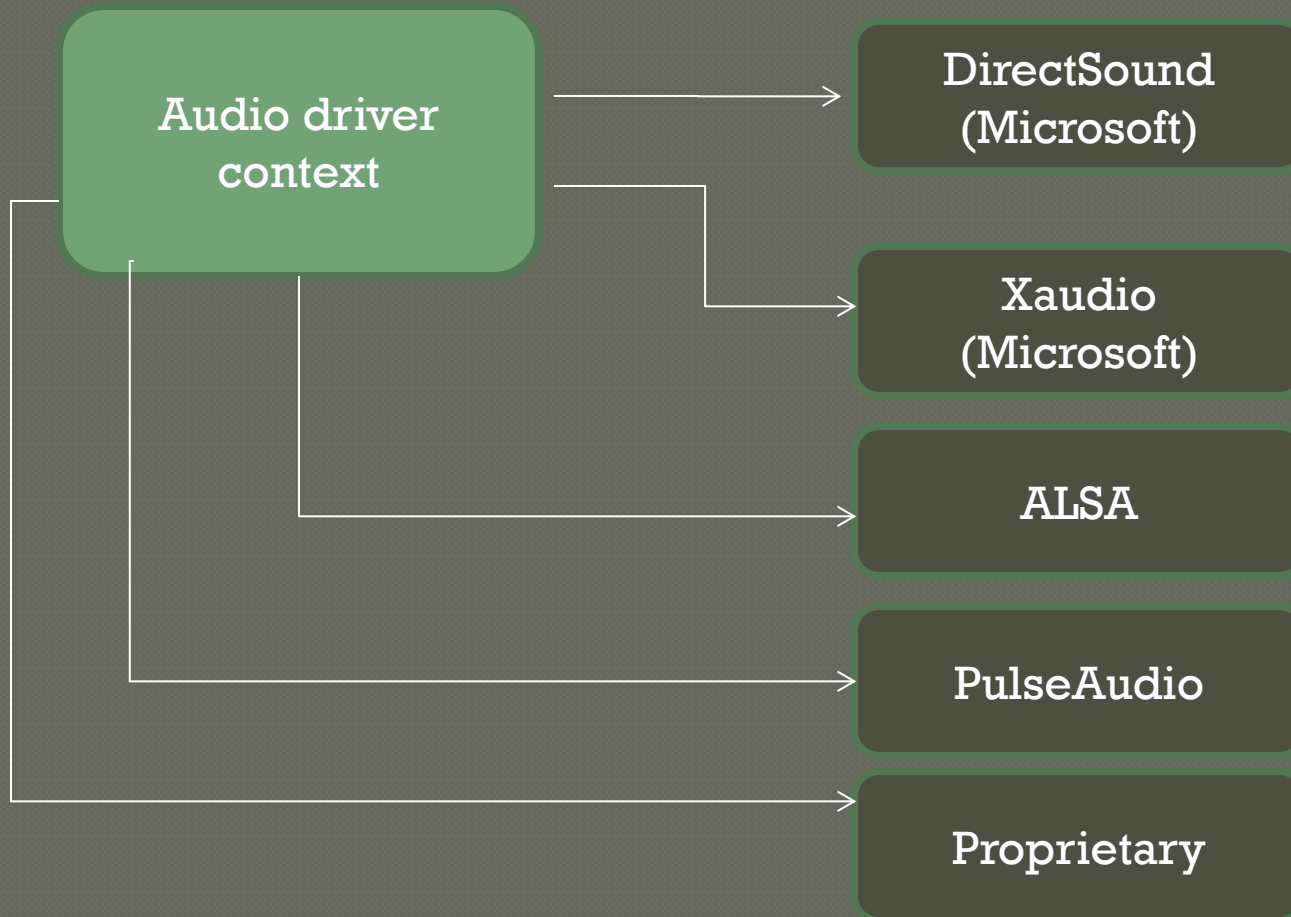


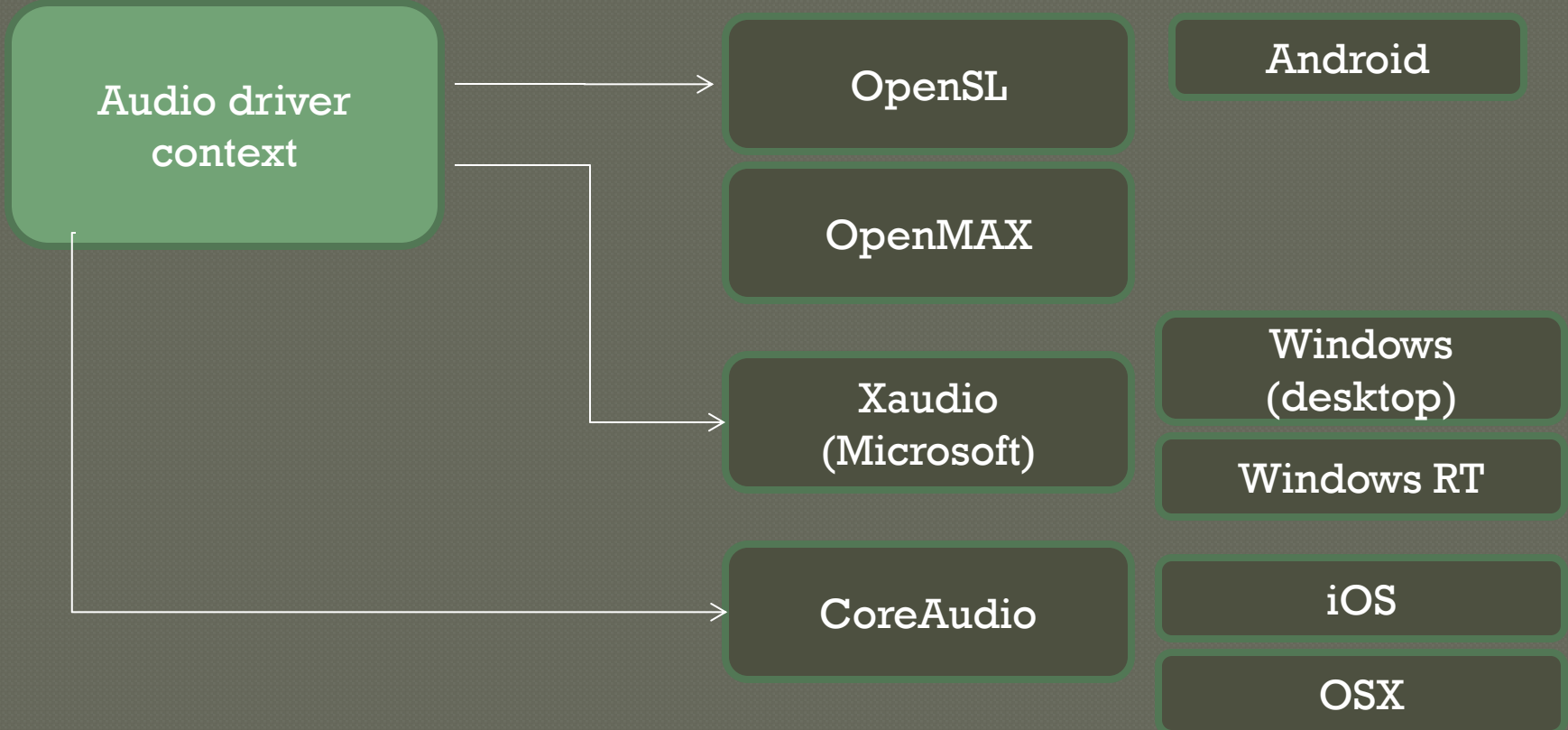
RetroArch





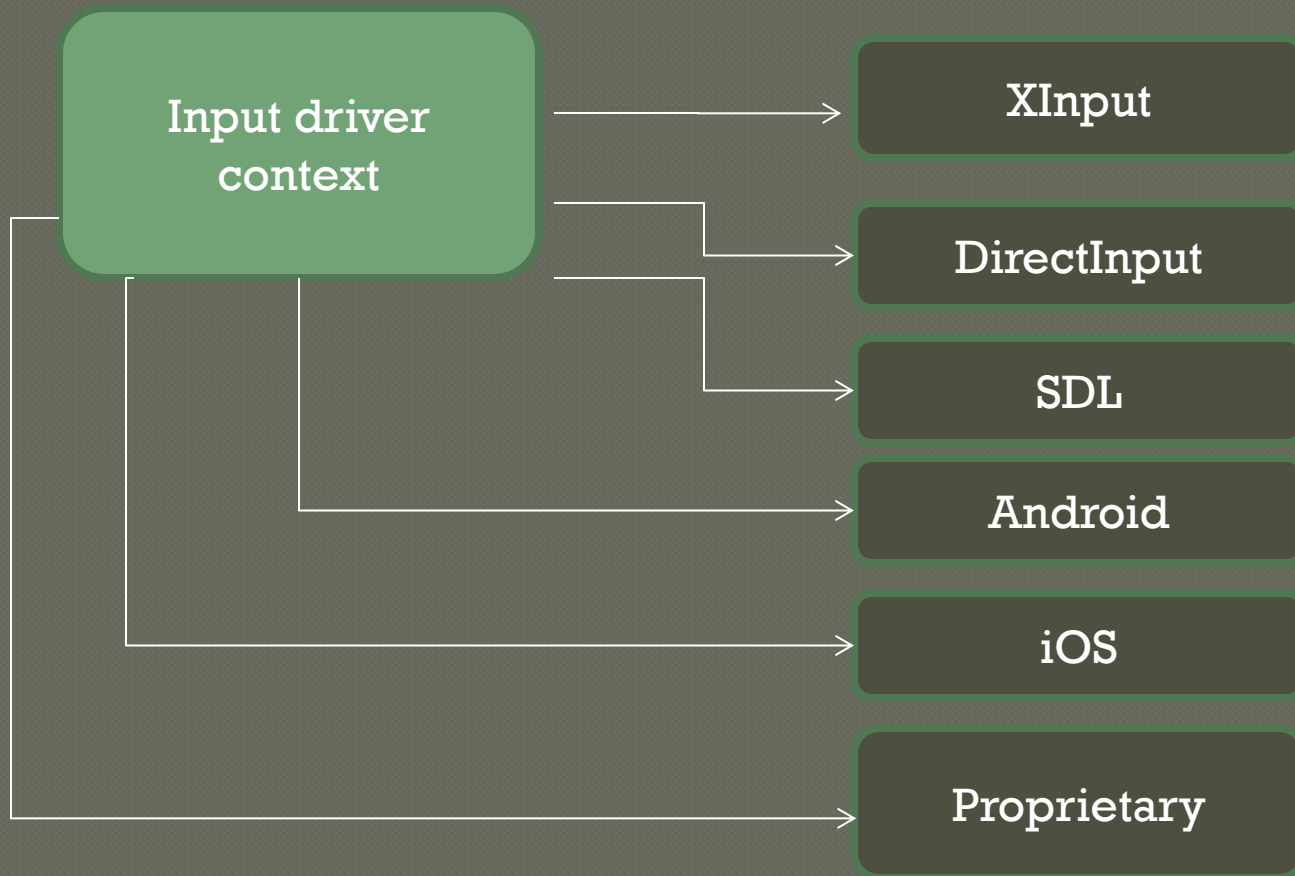
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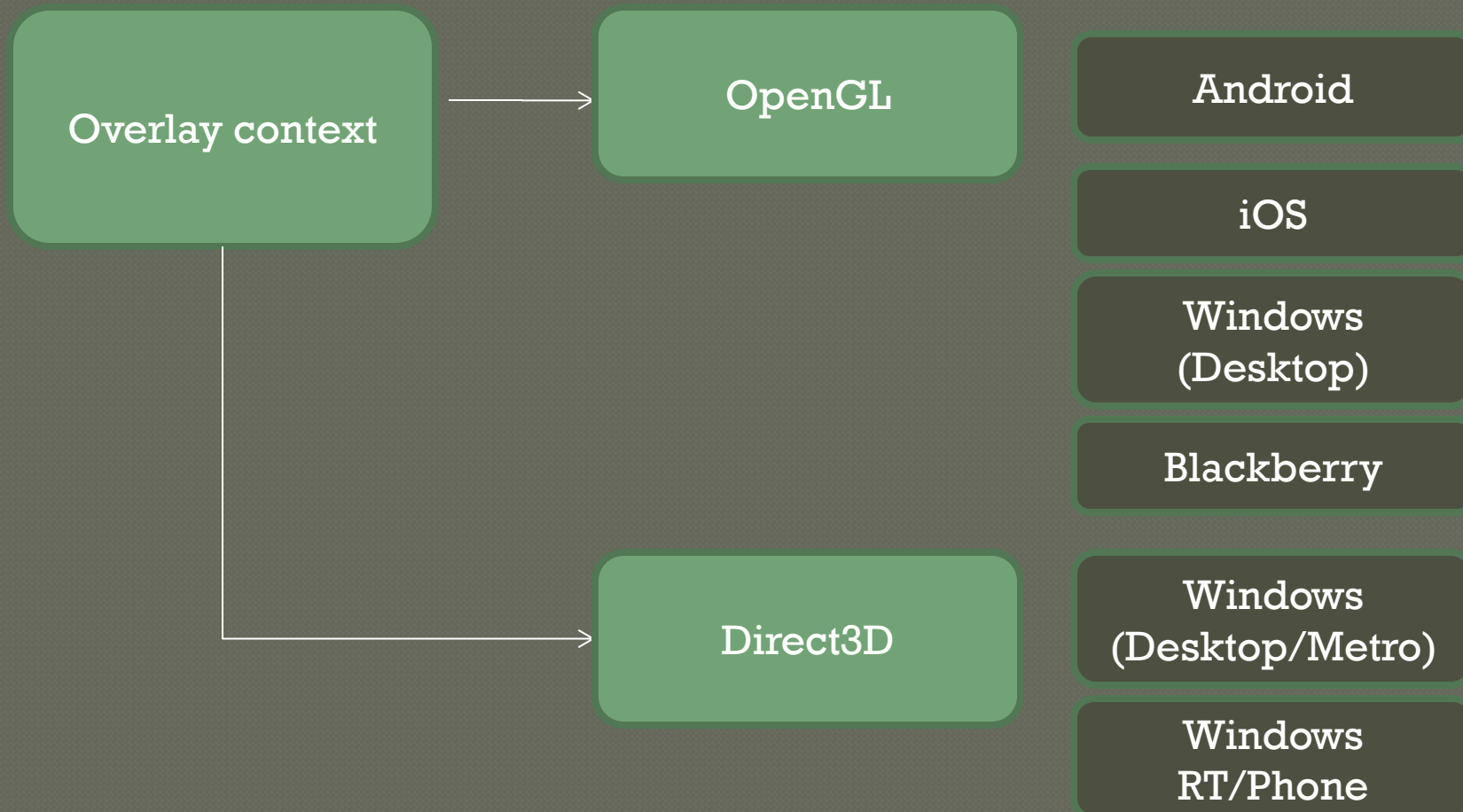


RetroArch



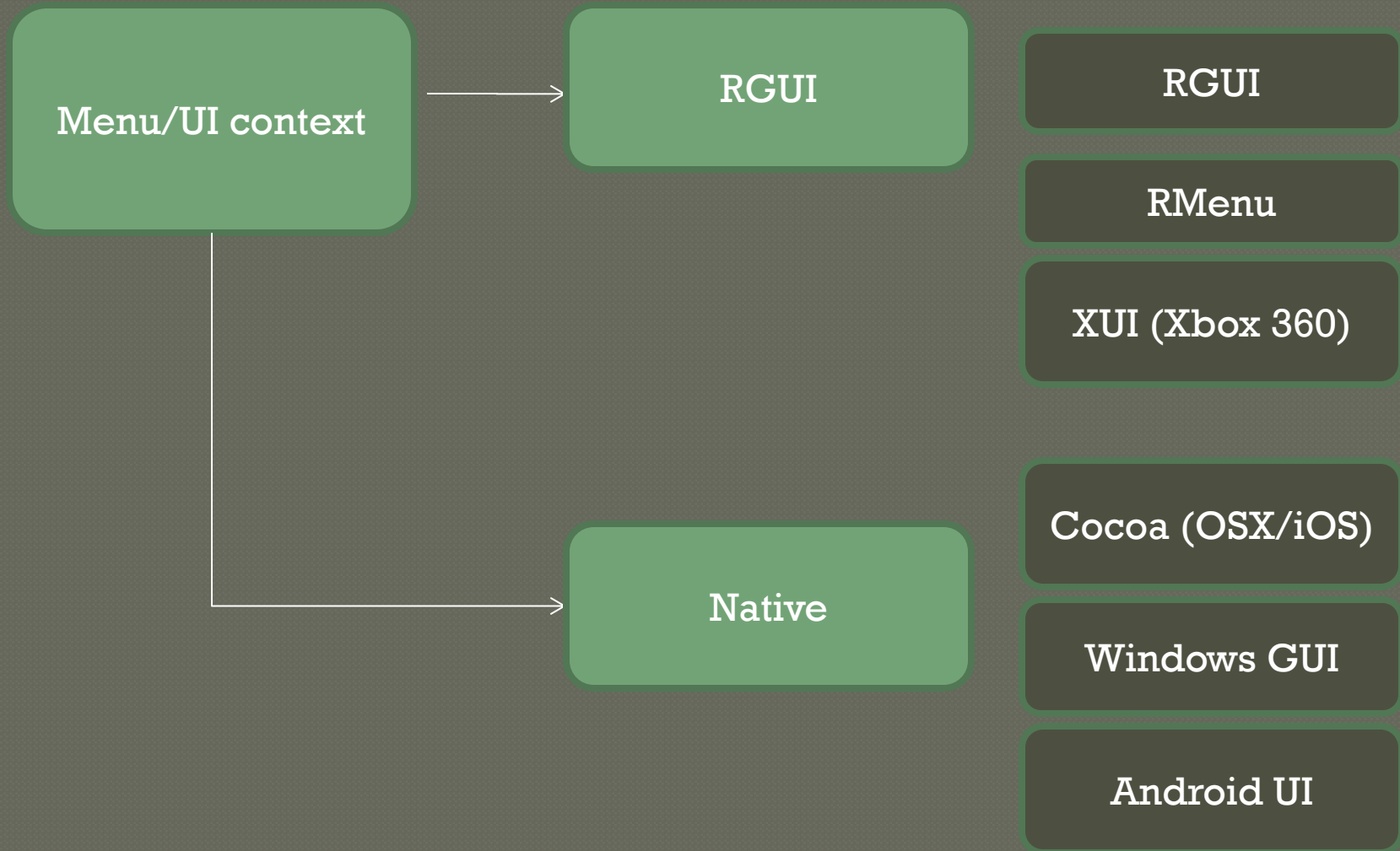


RetroArch





RetroArch



- Has its own software library of apps
 - Made possible by libretro API
 - Collection:
 - Games
 - Emulators
 - Media players
 - 3D apps
 - Augmented reality



● Usecase

- Playing games
 - Game ports
 - Emulators
- Movies/music/media
 - Media player implementations
- Streaming
- Augmented reality

● Userbase

- Mobile (Android/iOS)
- PC (Windows/OSX)
- Game consoles
- Embedded





Roadmap

- What RetroArch/libretro is trying to achieve
 - A platform / ecosystem made available on top of all other pre-existing platforms
 - A real-time, open-source alternative for rapid-deployment, cross-platform development
 - Non-game/emulator usecases
 - Gaming is going to meet CAD, home automation, augmented reality, and more
 - A bare-bones, non-restrictive alternative to more restrictive would-be competitors
 - Is already in use worldwide, now up to 300,000 hits on Google Play Store)

◉ Extension to libretro

- 3D rendering
 - By way of OpenGL / OpenGL ES
 - Crossplatform abstraction layer – Libretro GL
 - Is OpenGL not already crossplatform?



- What do most platforms have in common nowadays?



Allows us to write 3D graphics-based applications -

- In a platform-agnostic way
- Hardware-accelerated
- A universal language/API used worldwide by CAD developers, game developers, app developers, etc.

• What do these platforms NOT have in common?

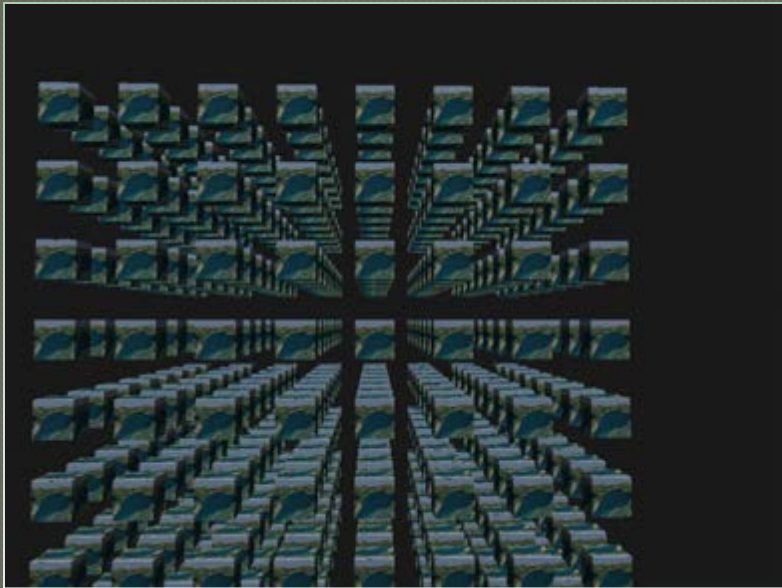
- Audio
- Input
- Shader
- Windowing implementations
- User interfaces
- Touchscreen overlays
- Camera
- Sensors
- Development environments
- And more...



- What is not-so portable about OpenGL?
 - Symbol wrapper lookup (necessary on Windows)
 - Divergent subsets of API functionality (GL ES 1/2/3, GL 1.5/2/3/4)
 - Windowing interfacing context drivers per platform
 - Display frontend for each platform
 - Post-processing by way of shaders
- What does Libretro GL provide?
 - A solution to all this and more...

RETRO ARCH

App World





- Continuously growing app library
- 47+ libretro cores
 - C/C++ codebases
 - Crossplatform

◉ Gifts

- Hardware products and accessories
- For the purposes of porting RetroArch to it, enhancing support, features, etc.



- Also on the web
 - Every app from the libretro ecosystem available on a web-browser
 - Using Javascript
- Webrowsers
 - New platforms in their own right
 - Every platform has its own Javascript implementation
- Recompiling C/C++ code into Javascript
 - Emscripten
 - Native apps running inside webbrowser with decent performance



- Windows RT/Phone support
 - Being everywhere
 - On any device
- A big online services presence
 - Matchmaking
 - Chatlobby's
 - Live streaming
 - Augmented reality services
 - Etc





Crossplatform architecture

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